b. Process Air Compressor (K-302)

Atomospheric air passes through an Air Intake Stack (H-301) and Filter (F-301) to separate sand, dust, etc. and enters the first stage of the Process Air Compressor. The compressed air from an intermediate point at the first case

flows to the First Stage Air Cooler (E-305) where heat developed by compression is removed and any water condensed is knocked out in the First Stage Air Separator (V-305). The condensed water is drawn off and discharged to sewer. Then the process air flows into the remaining stages and cases of the Process Air Compressor. Process air is further compressed in each stage and discharged to its respective Interstage Cooler where compression heat is removed in water cooled exchangers. Water is condensed, separated in each Interstage Separator, drawn off and discarded to sewer. Before entering the fourth stage of the Compressor, a part of the air is withdrawn for use in the Urea Unit, for addition to the Catacarb solution and for instrument air requirements. The air required as instrument air is sent to the Instrument Air Dryer. The air required for process then enters the fourth stage of Air Compressor to be compressed to about 34 Kg/cm^2G, and flows to the Gas Reform Section. The discharge pressure is maintained by control of the Turbine speed. The flow required to avoid surge, as measured at the discharge, is maintained by venting air to the atmosphere.

c. Natural Gas Compressor (K-303)

Natural gas from the battery limit, passes through the Natural Gas Drain Separator (V-111) and enters the Natural Gas Compressor. Natural gas is compressed to 40 Kg/cm^2G, and flows to the Gas Reform Section.

The discharge pressure is maintained by control of the Turbine speed. The flow required to avoid surge, as measured at the suction, is maintained by returning natural gas from the Compressor discharge through the Natural Gas By-Pass Cooler (E-308).